English

WIRESHARK

NETWORK TRAFFIC ANALYSIS REPORT

MF0487_3: Information Security Auditing

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Teacher: Miguel Hernández

YOUR TASK

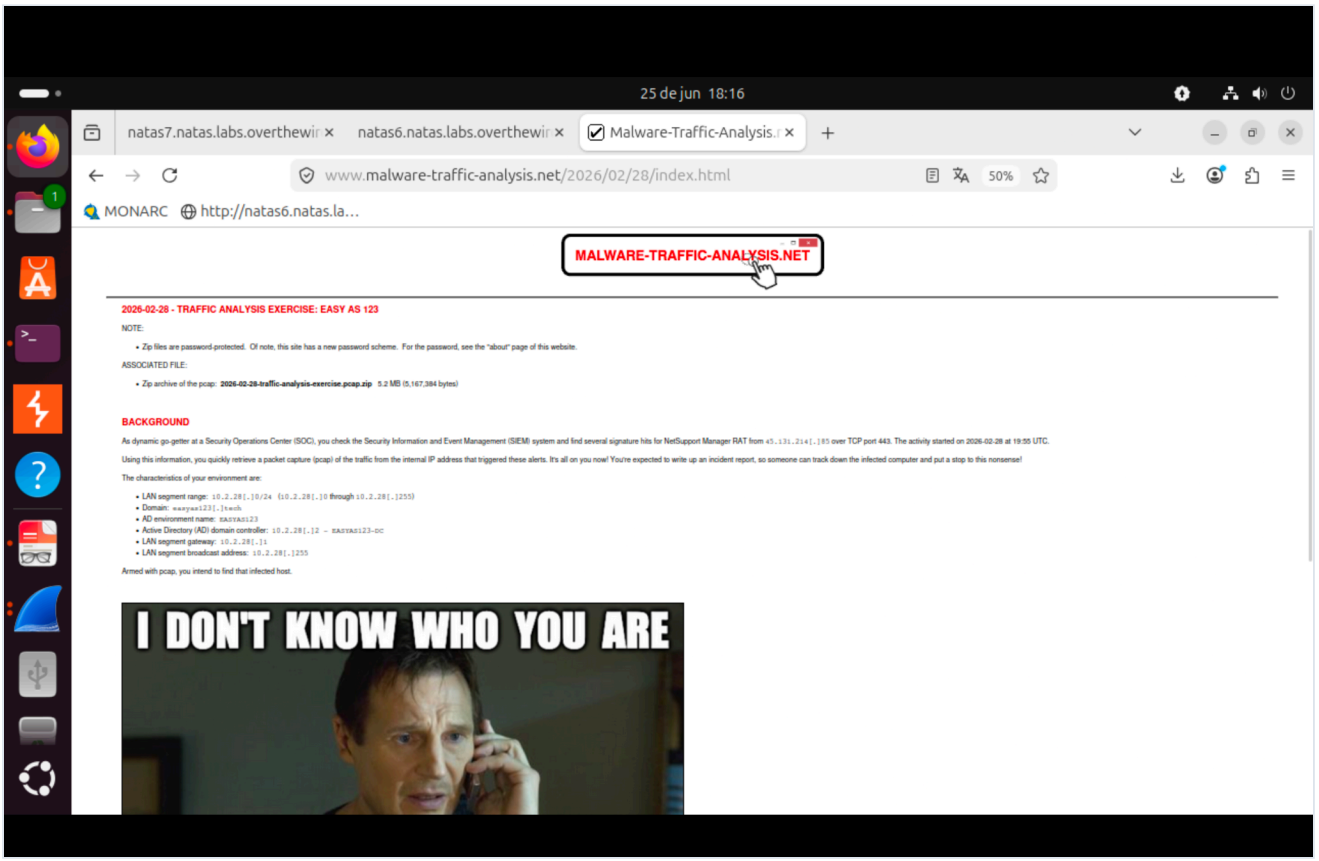
For this exercise, answer the following questions for your incident report:

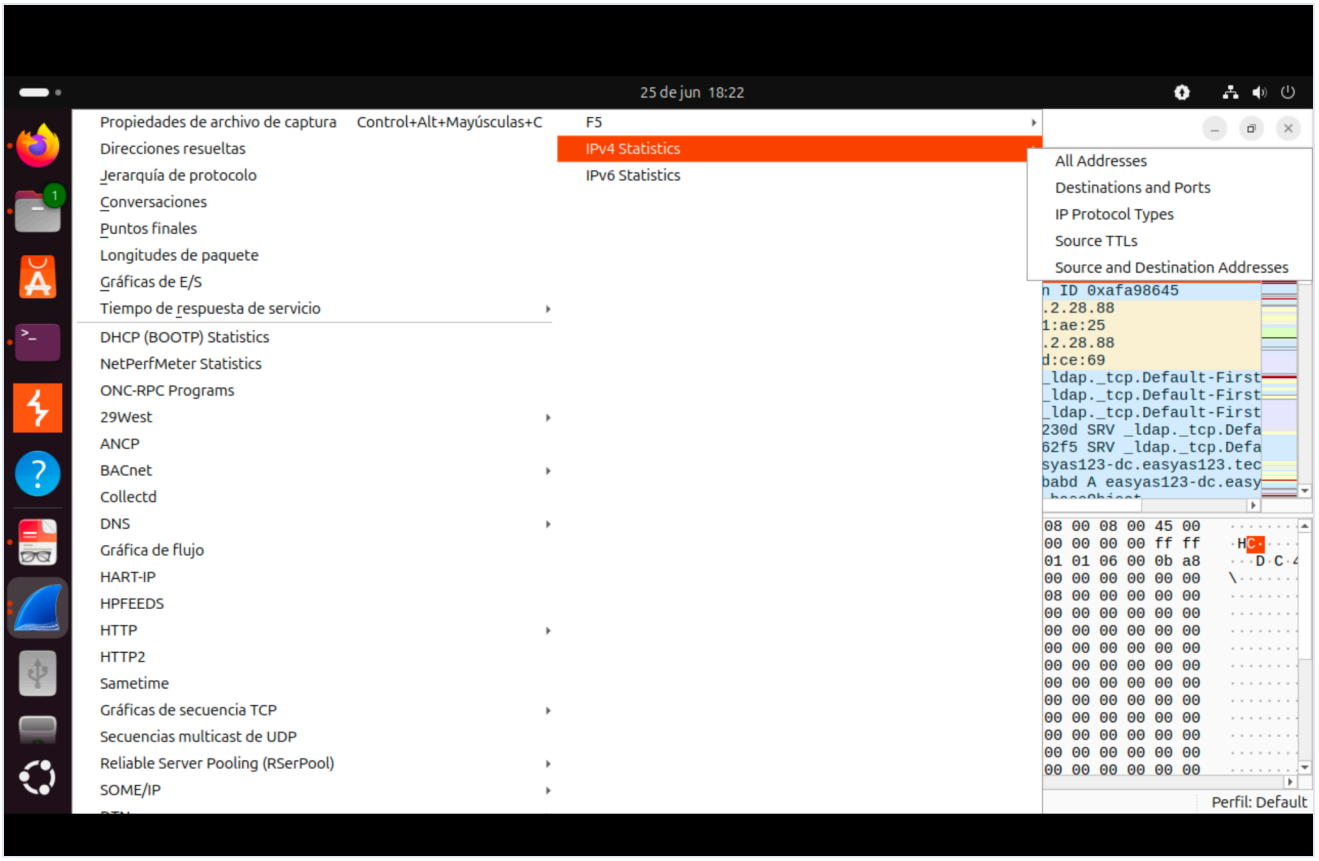
- What is the IP address of the infected Windows client?
- What is the MAC address of the infected Windows client?
- What is the hostname of the infected Windows client?
- What is the user account name on the infected Windows client?
- What is the full name of the user associated with the account?

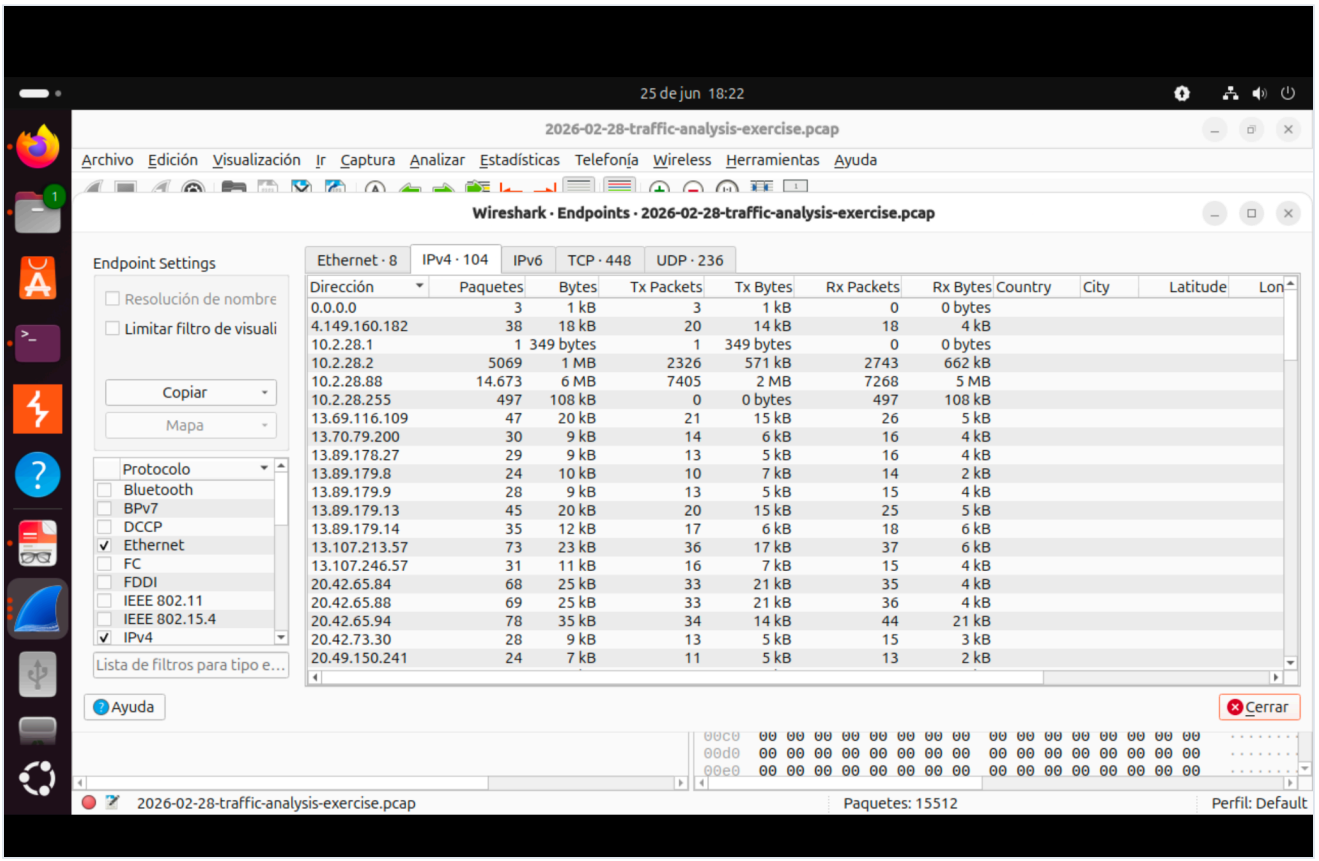
SU TAREA

Para este ejercicio, responda las siguientes preguntas para su informe de incidente:

- ¿Cuál es la dirección IP del cliente de Windows infectado?
- ¿Cuál es la dirección MAC del cliente de Windows infectado?
- ¿Cuál es el nombre del host del cliente de Windows infectado?
- ¿Cuál es el nombre de la cuenta de usuario del cliente de Windows infectado?
- ¿Cuál es el nombre completo del usuario de la cuenta de usuario?



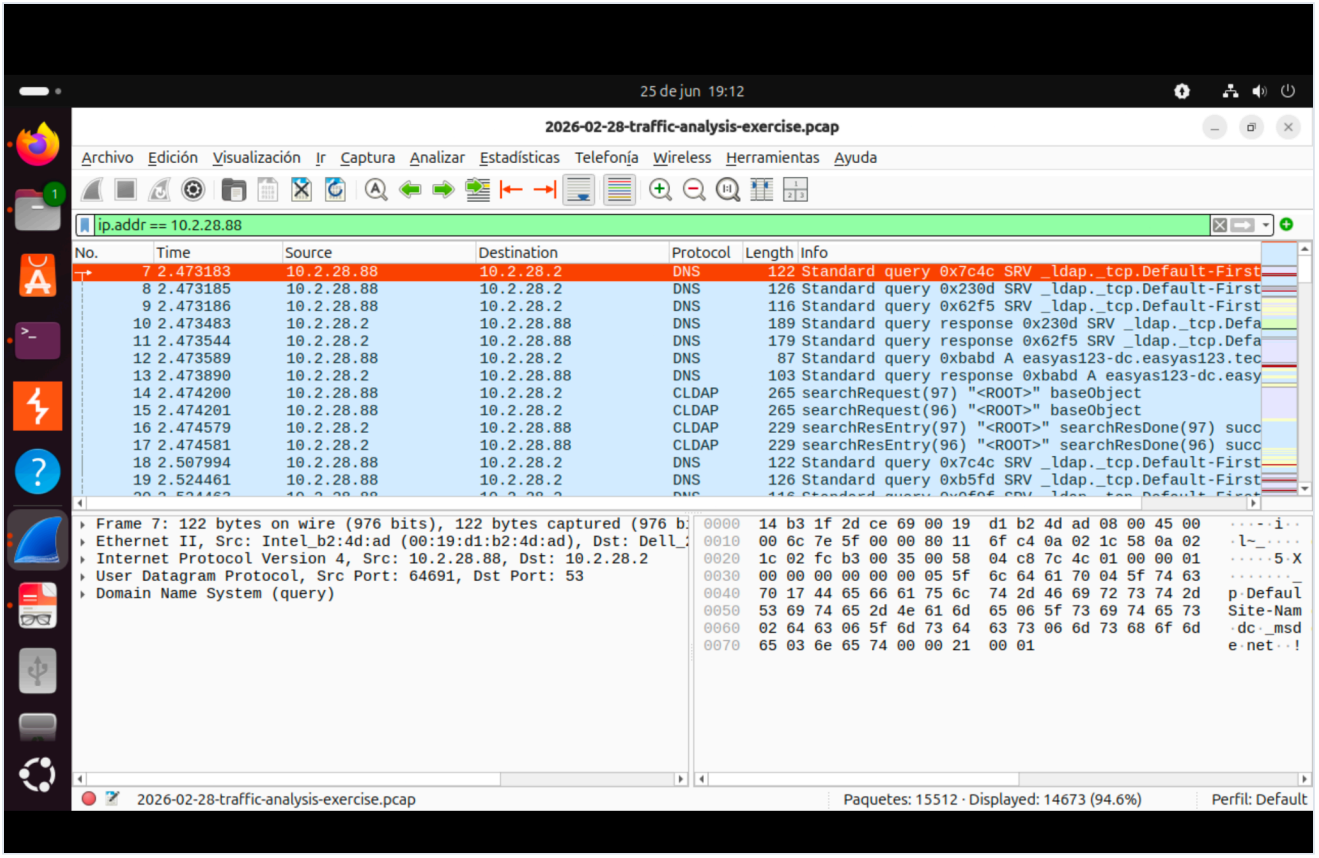

25 de jun 18:22
Propiedades de archivo de captura Control+Alt+Mayúsculas+C
Direcciones resueltas
Jerarquía de protocolo
Conversaciones
Puntos finales
Longitudes de paquete
Gráficas de E/S
Tiempo de respuesta de servicio
DHCP (BOOTP) Statistics
NetPerfMeter Statistics
ONC-RPC Programs
29West
ANCP
BACnet
Collectd
DNS
Gráfica de flujo
HART-IP
HPFEEDS
HTTP
HTTP2
Sametime
Gráficas de secuencia TCP
Secuencias multicast de UDP
Reliable Server Pooling (RSerPool)
SOME/IP
F5
IPv4 Statistics
IPv6 Statistics
All Addresses
Destinations and Ports
IP Protocol Types
Source TTLs
Source and Destination Addresses
Perfil: Default


25 de jun 18:22
2026-02-28-traffic-analysis-exercise.pcap
Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda
Wireshark · Endpoints · 2026-02-28-traffic-analysis-exercise.pcap
Endpoint Settings
Resolución de nombre
Limitar filtro de visuali
Copiar
Mapa
Protocolo
Bluetooth
BPv7
DCCP
Ethernet
FC
FDDI
IEEE 802.11
IEEE 802.15.4
IPv4
Lista de filtros para tipo e...
Ethernet · 8
IPv4 · 104
IPv6
TCP · 448
UDP · 236
Dirección Paquetes Bytes Tx Packets Tx Bytes Rx Packets Rx Bytes Country City Latitude Lon
0.0.0.0 3 1 kB 3 1 kB 0 0 bytes
4.149.160.182 38 18 kB 20 14 kB 18 4 kB
10.2.28.1 1 349 bytes 1 349 bytes 0 0 bytes
10.2.28.2 5069 1 MB 2326 571 kB 2743 662 kB
10.2.28.88 14.673 6 MB 7405 2 MB 7268 5 MB
10.2.28.255 497 108 kB 0 0 bytes 497 108 kB
13.69.116.109 47 20 kB 21 15 kB 26 5 kB
13.70.79.200 30 9 kB 14 6 kB 16 4 kB
13.89.178.27 29 9 kB 13 5 kB 16 4 kB
13.89.179.8 24 10 kB 10 7 kB 14 2 kB
13.89.179.9 28 9 kB 13 5 kB 15 4 kB
13.89.179.13 45 20 kB 20 15 kB 25 5 kB
13.89.179.14 35 12 kB 17 6 kB 18 6 kB
13.107.213.57 73 23 kB 36 17 kB 37 6 kB
13.107.246.57 31 11 kB 16 7 kB 15 4 kB
20.42.65.84 68 25 kB 33 21 kB 35 4 kB
20.42.65.88 69 25 kB 33 21 kB 36 4 kB
20.42.65.94 78 35 kB 34 14 kB 44 21 kB
20.42.73.30 28 9 kB 13 5 kB 15 3 kB
20.49.150.241 24 7 kB 11 5 kB 13 2 kB
Ayuda
Cerrar
2026-02-28-traffic-analysis-exercise.pcap
Paquetes: 15512
Perfil: Default

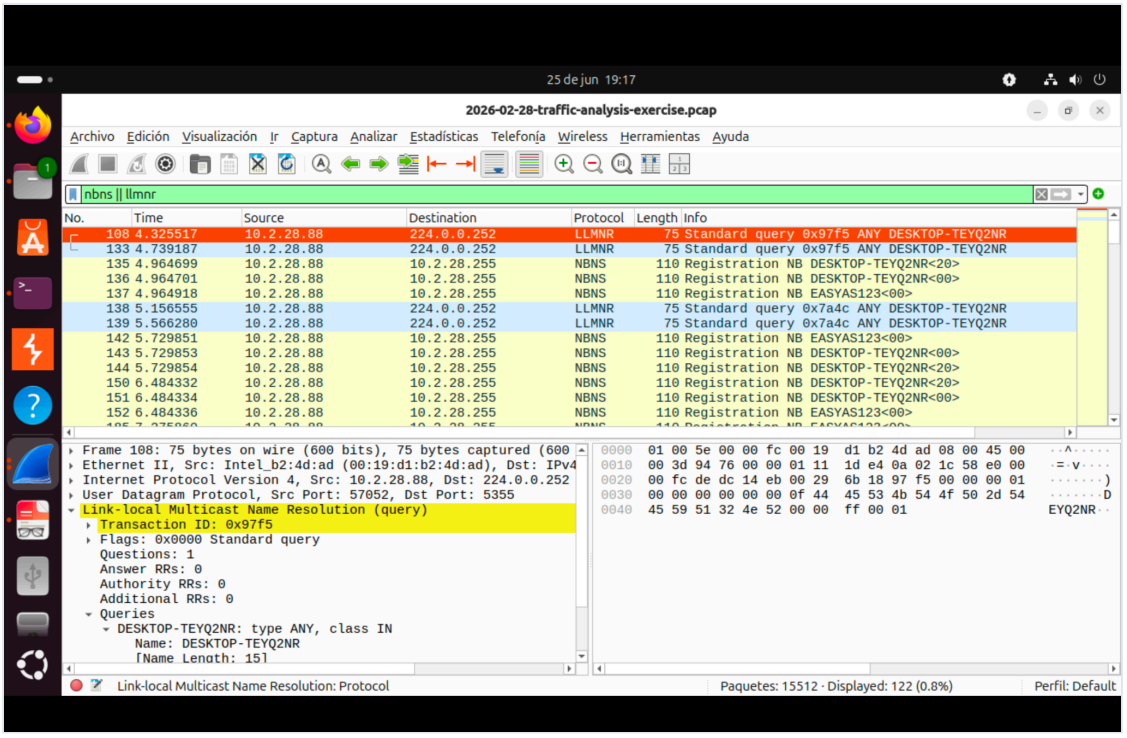
During the initial analysis of the network traffic capture using Wireshark, the Endpoints list was reviewed to identify the devices involved in the communication. The IP address 10.2.28.88 was observed to account for the highest traffic volume among devices on the internal network, so it was considered the main suspect device and the starting point for further investigation.

Using this IP address, we then searched for additional information, such as the MAC address, hostname and user details, by analysing the name resolution and authentication protocols present in the capture.

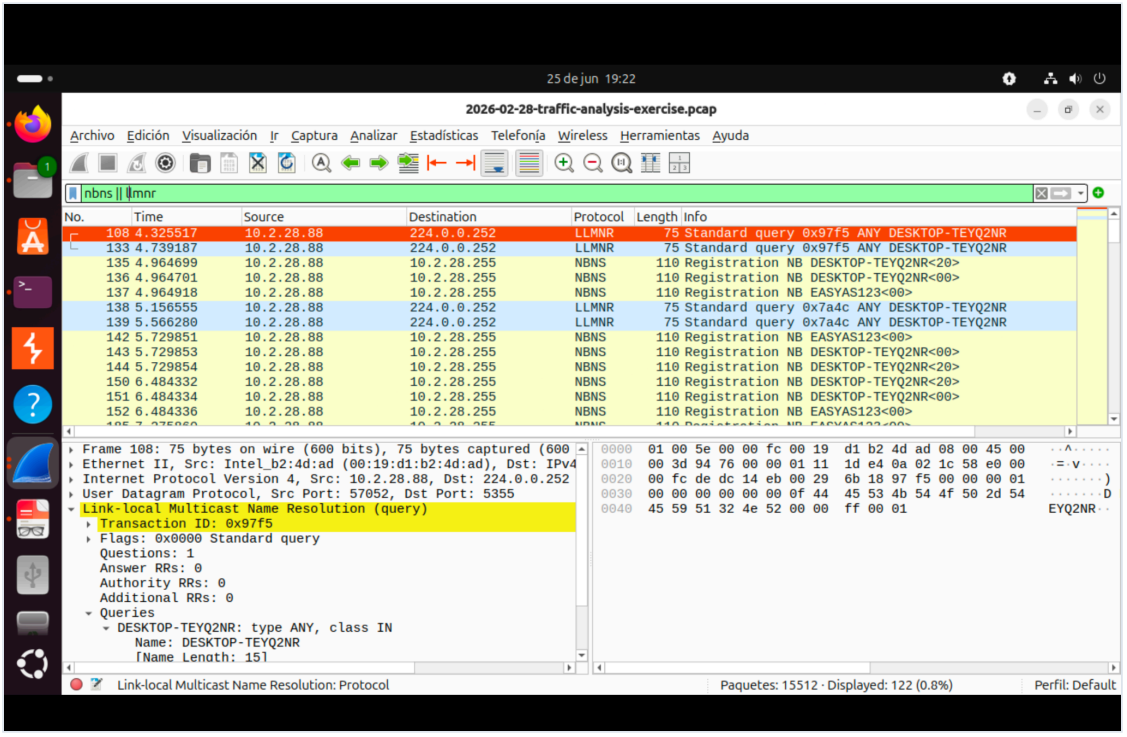
We apply a filter to see what is happening at this IP address:



> Before treating 10.2.28.88 as suspicious, we need to find evidence to support that suspicion (for example, that it communicates with the malware server or generates traffic related to NetSupport Manager RAT).



We filter by the LLMNR(?)[1] protocol.



[1] We filter by these protocols because each one usually contains a different type of information. It is like knowing which folder to look in for a document.

- LLMNR → Similar to NBNS; it can also reveal the computer name.
- Kerberos → This is the Windows authentication protocol. When a user logs on to a domain, the username and domain often appear here.

Think of yourself as a detective. If you want to find out a person's name, you do not look in the traffic fines register; you go to the civil registry. The same applies here: to find the user, you examine authentication protocols (Kerberos or NTLM). To find the computer name, you examine name resolution protocols (NBNS or LLMNR). This is not a random filter: it is chosen because that protocol normally carries the information you need to find.

We look at the Info column.

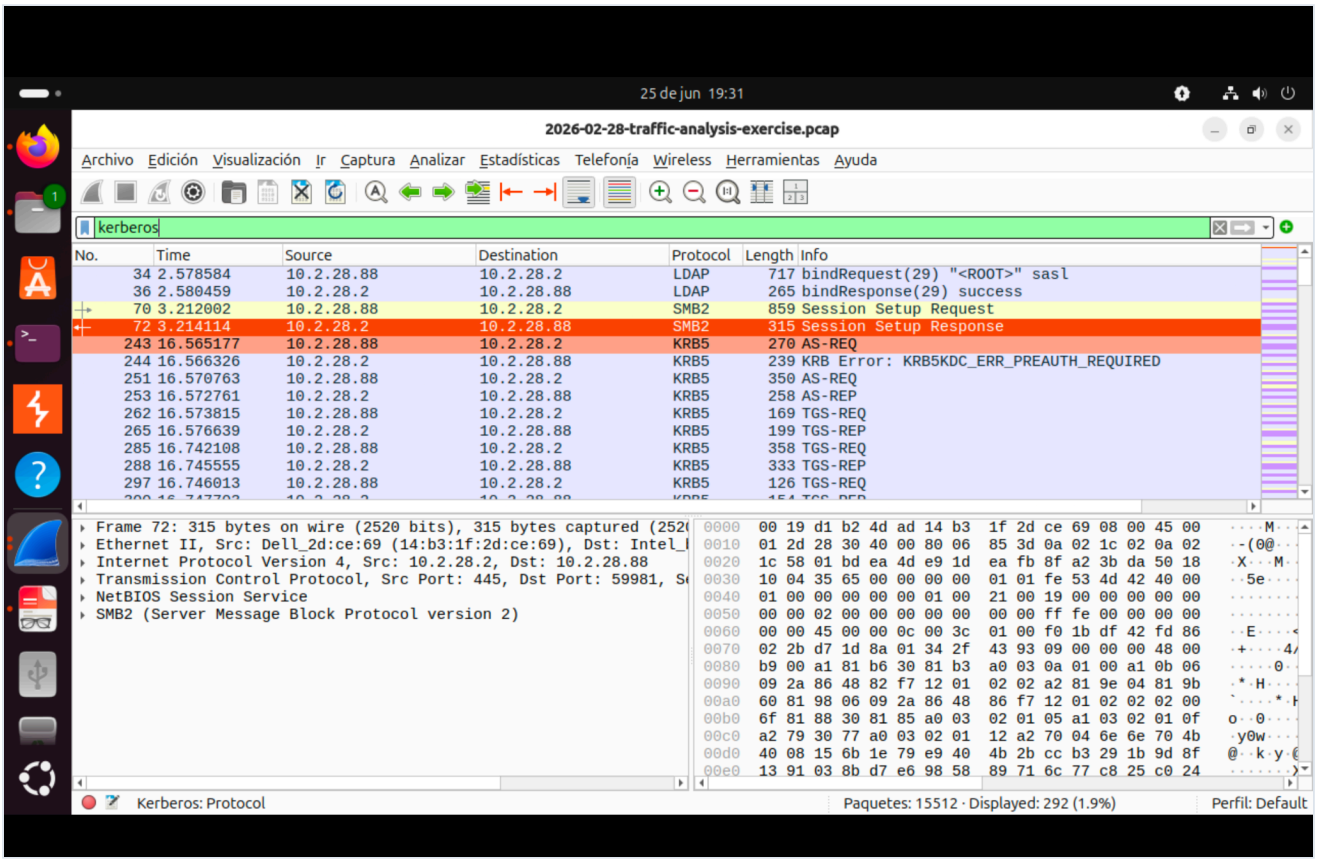
No.	Time	Source	Destination	Protocol	Length	Info
108	4.325517	10.2.28.88	224.0.0.252	LLMNR	75	Standard query 0x97f5 ANY DESKTOP-TEYQ2NR

This means that the hostname (computer name) is:

```
DESKTOP-TEYQ2NR
```

We have also confirmed that these requests come from IP address 10.2.28.88, so we now have two related pieces of information:

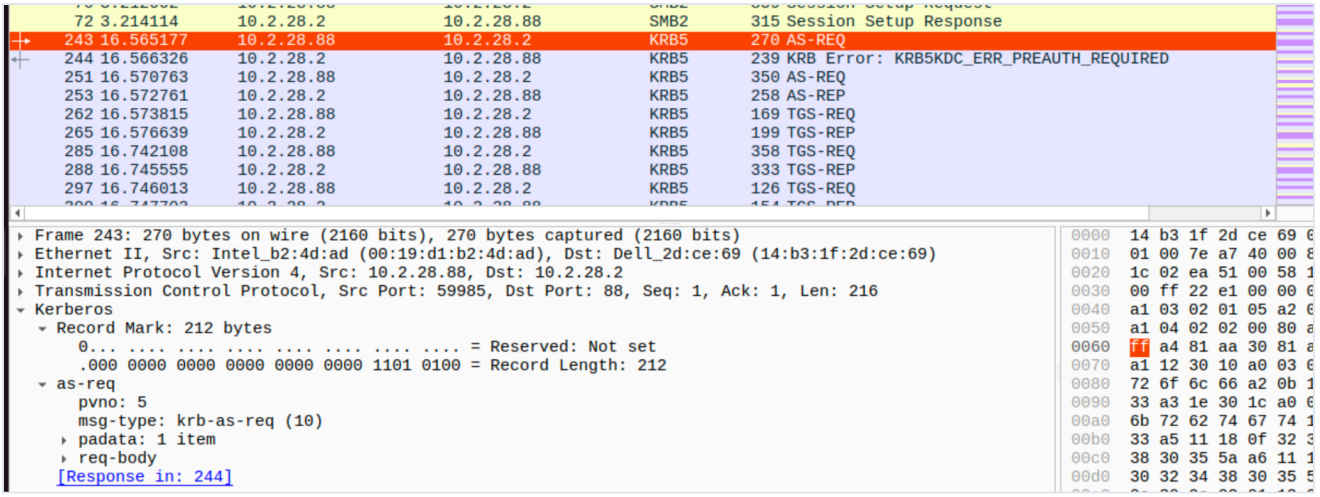
- IP: 10.2.28.88
- Hostname: DESKTOP-TEYQ2NR



We can see that the kerberos filter is returning traffic (KRB5), which is a good sign. However, the packet list still does not show the username, only authentication messages (AS-REQ, AS-REP, TGS-REQ, etc.).

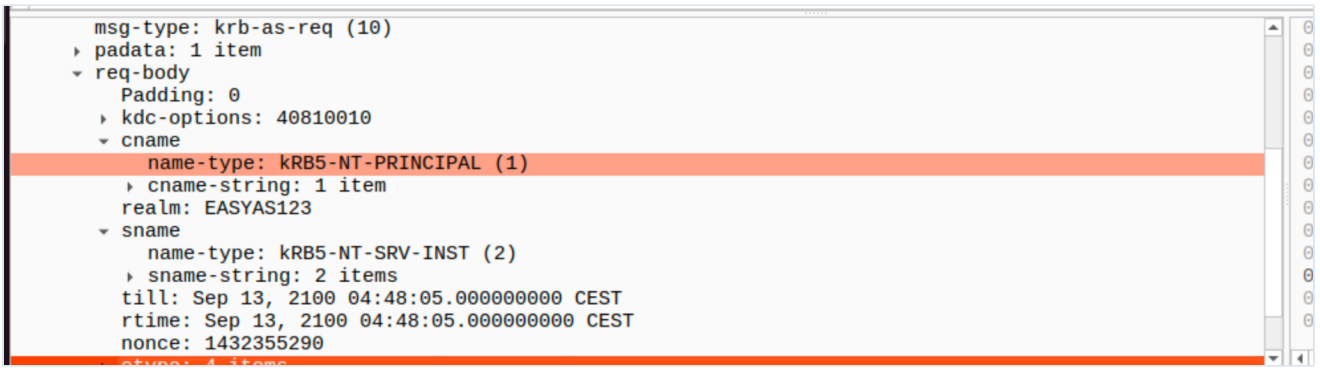
The next step:

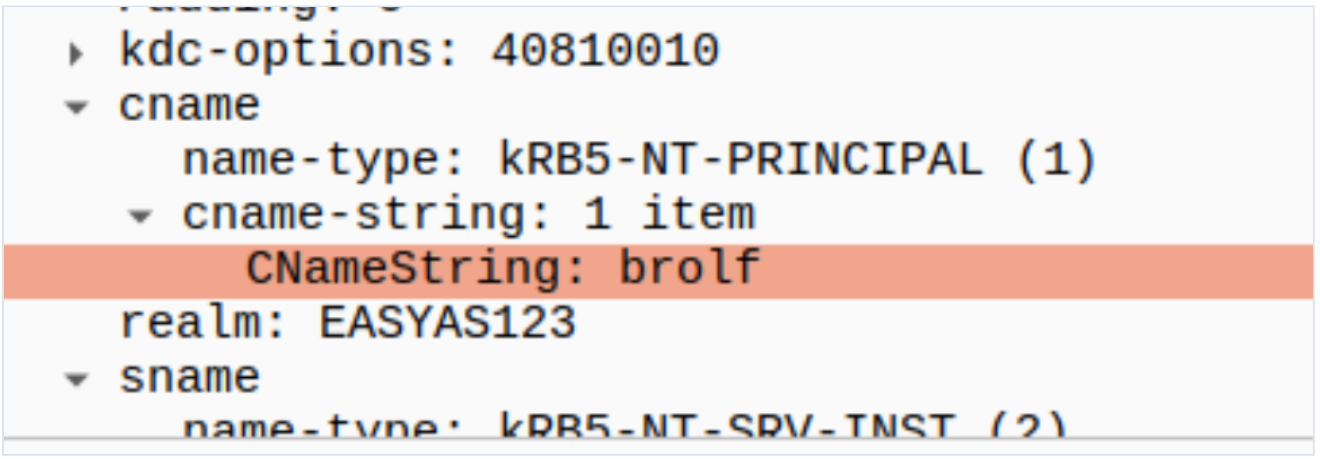
We right-click one of the AS-REQ packets (for example, no. 243) and select: Expand → Kerberos.



We can also see the MAC address in the previous screenshot:

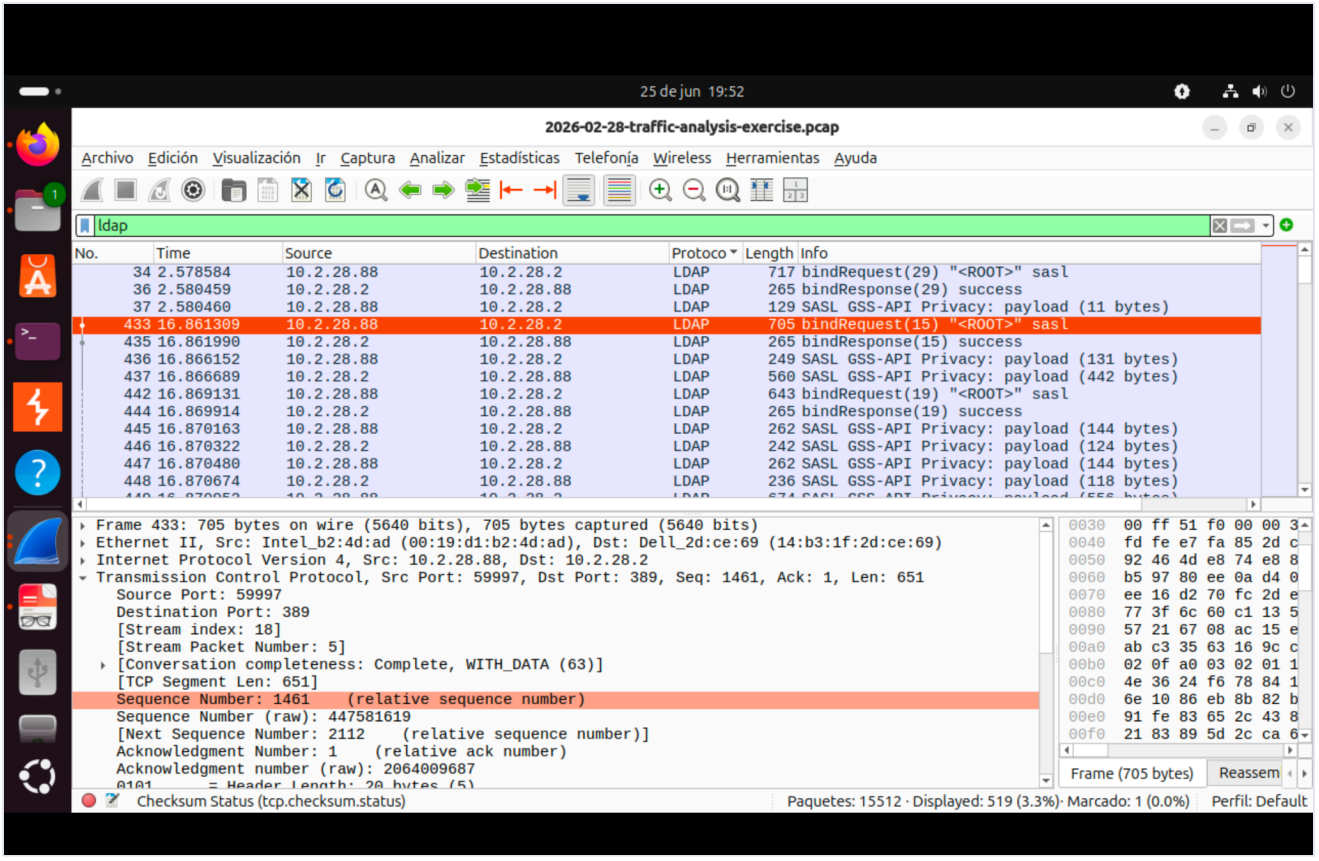
```
Ethernet II
Src: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)
```

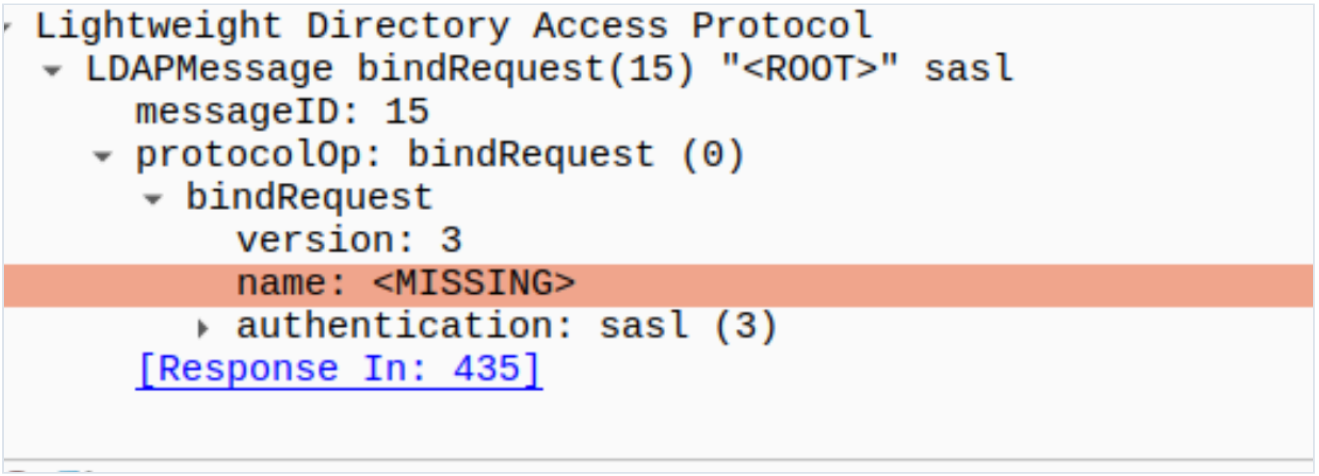




Answers we have so far

- IP address: 10.2.28.88
- MAC address: 00:19:d1:b2:4d:ad
- Hostname: DESKTOP-TEYQ2NR
- User account: brolf





That bindRequest will not give us the full name because it says:

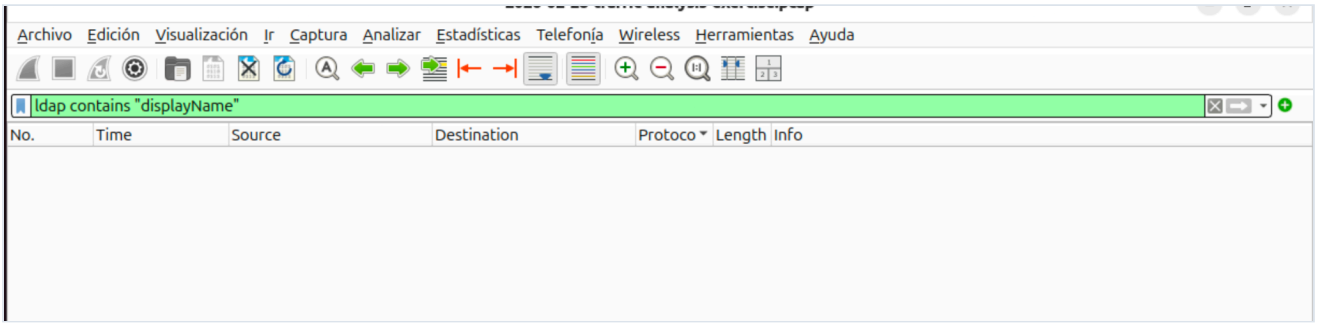
This means that authentication is being performed using SASL/Kerberos, and the username is not transmitted in plaintext within the LDAP packet.

What would an analyst do now?

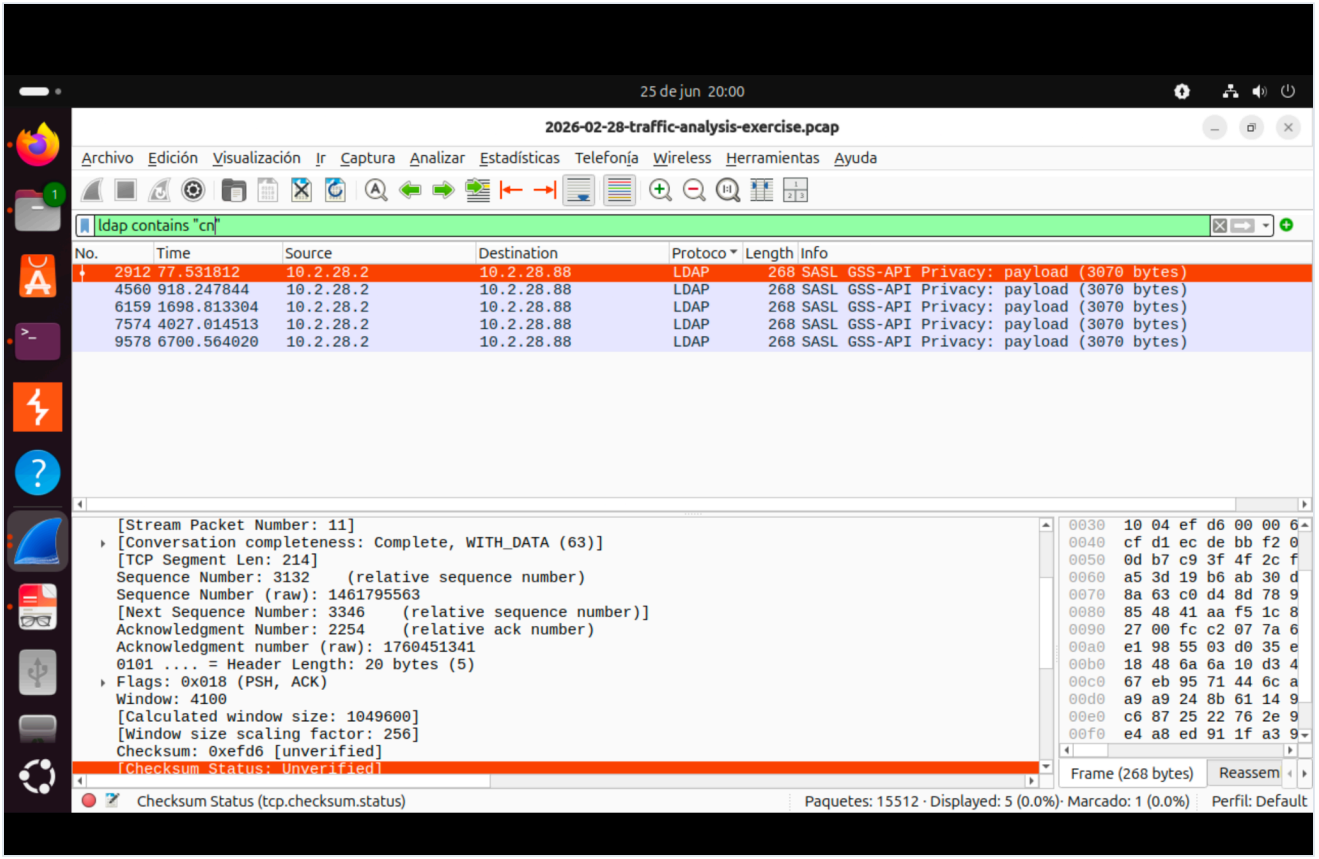
They would stop searching in that bindRequest, because they already know the information is not there. They would look elsewhere for a response in which Active Directory returns the user's attributes.

First, we try this filter:

```
ldap contains "displayName"
```



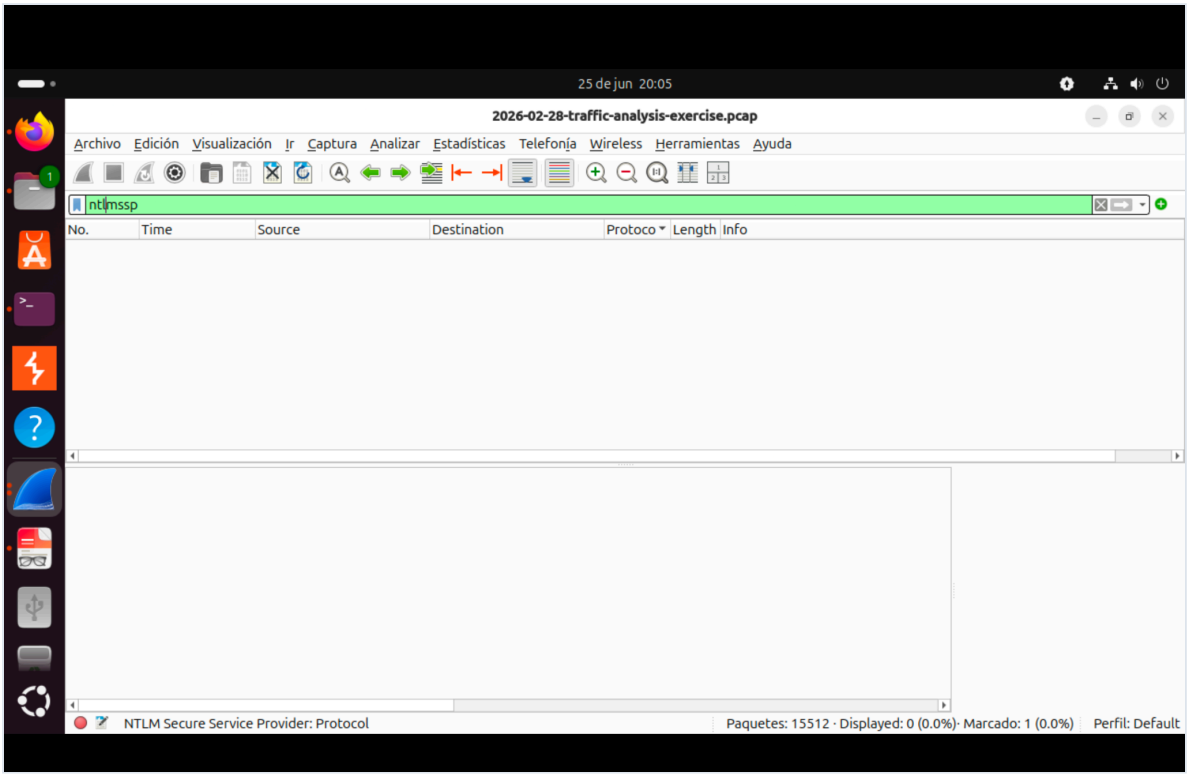
Nothing; it returns no results.

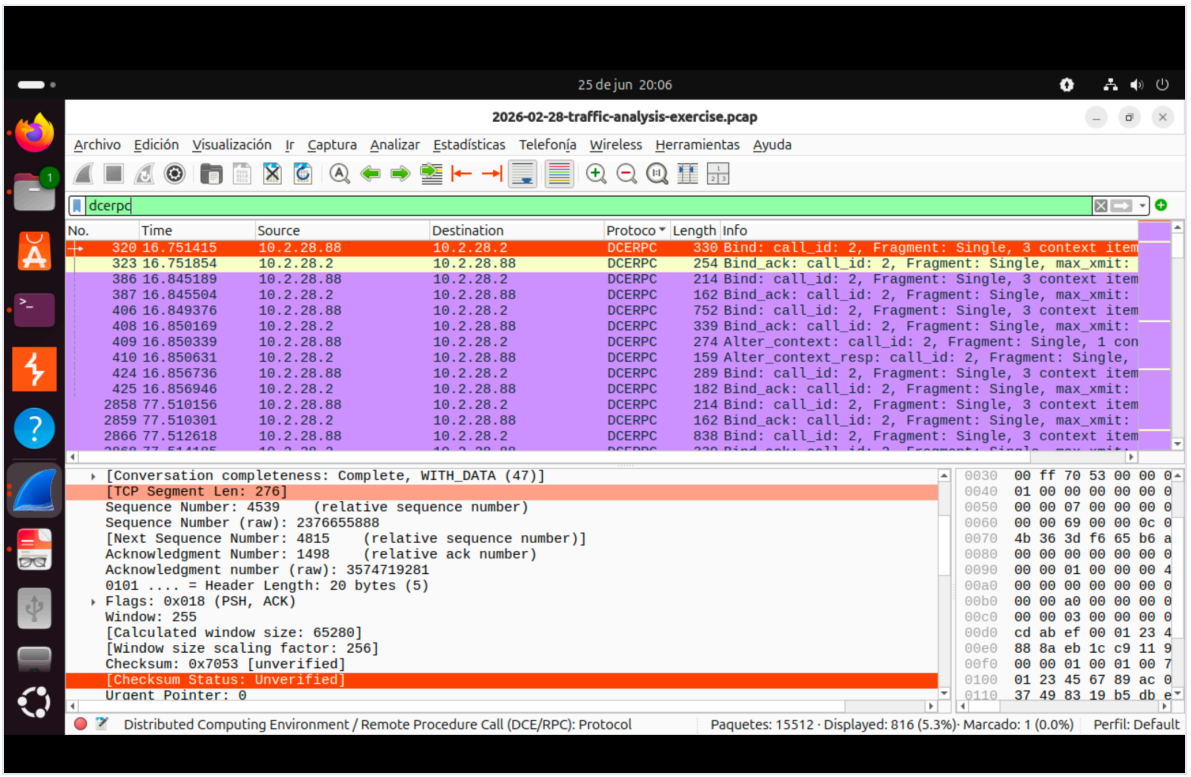


It only finds packets labelled:

```
SASL GSS-API Privacy
```

In other words, the LDAP content is encrypted. Wireshark finds the string “cn” within the stream, but it cannot decrypt the attributes (such as the user’s displayName or cn).



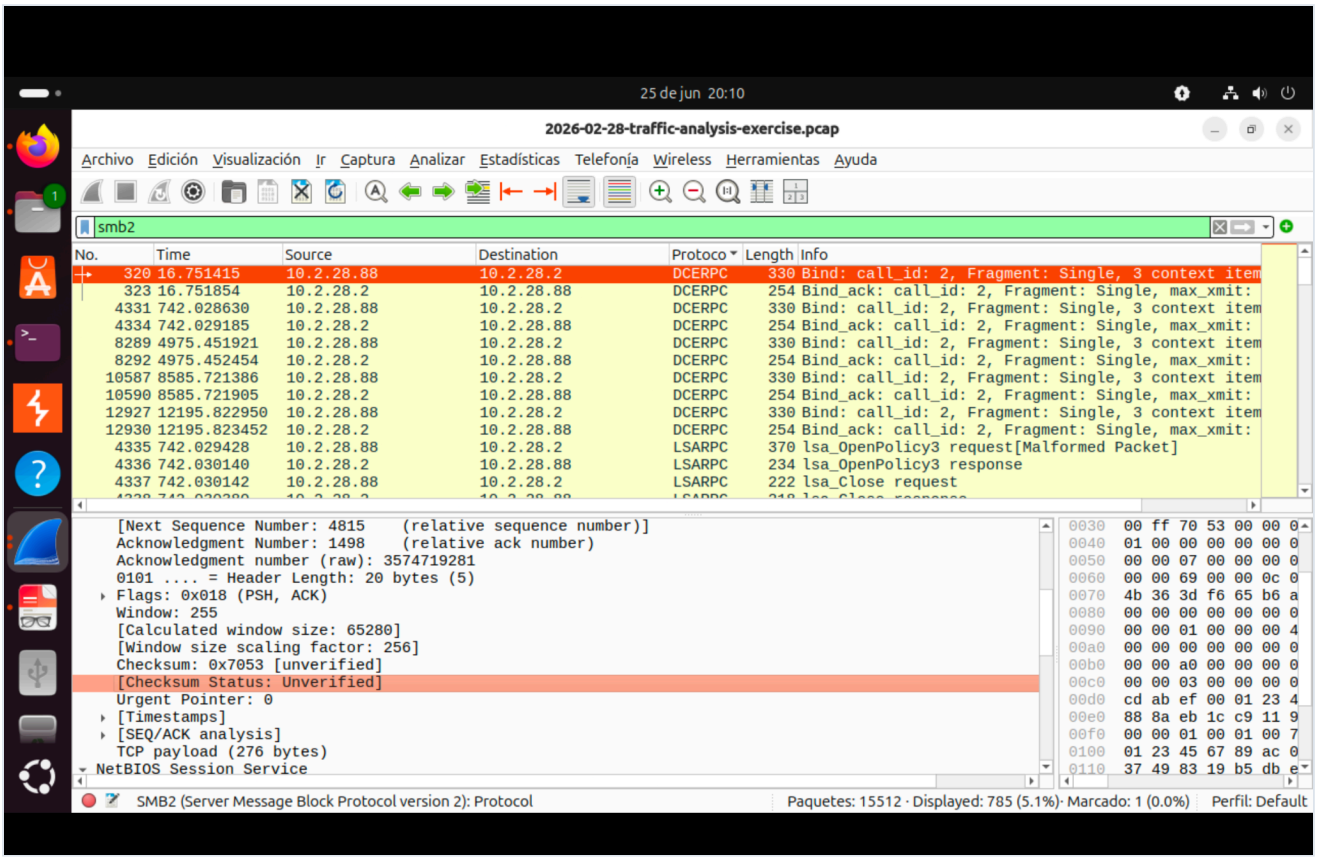


The full name is not visible here either. The packets shown as Bind and Bind_ack are only establishing communication; they do not contain the user's details.

The next step is much more promising:

We try these filters one at a time:

```
smb2
```



The capture shows LSARPC calls (Lsa_OpenPolicy3). This protocol is used to query Windows security information, including users and policies.

We try the following:

1. Click packet 4335 (LSARPC Lsa_OpenPolicy3 request).
2. In the lower pane, expand:

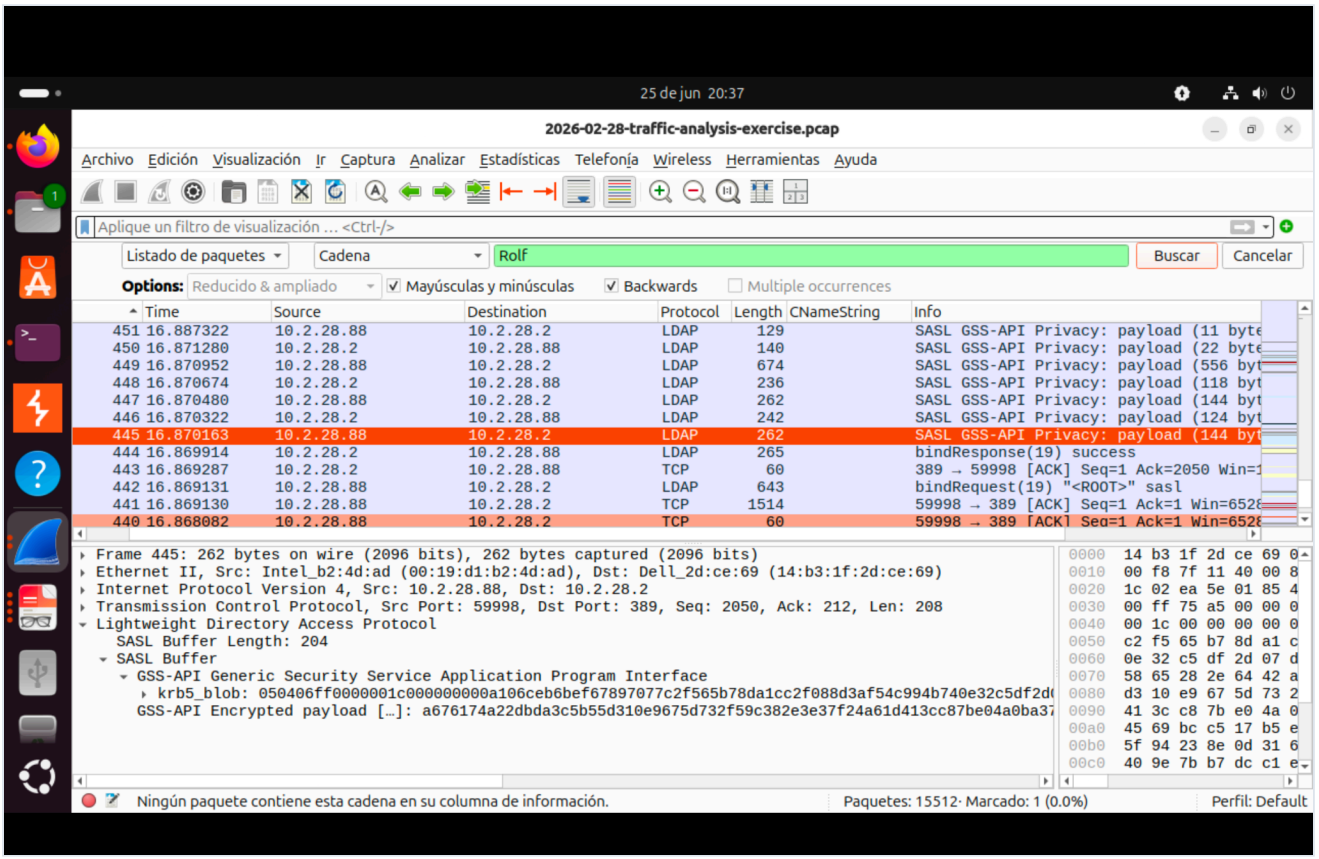
   Netlogon (if present)

   DCE/RPC

LSARPC

3. Look for fields named:

- Account Name
- User Name
- Display Name
- Full Name
- cn
- sAMAccountName



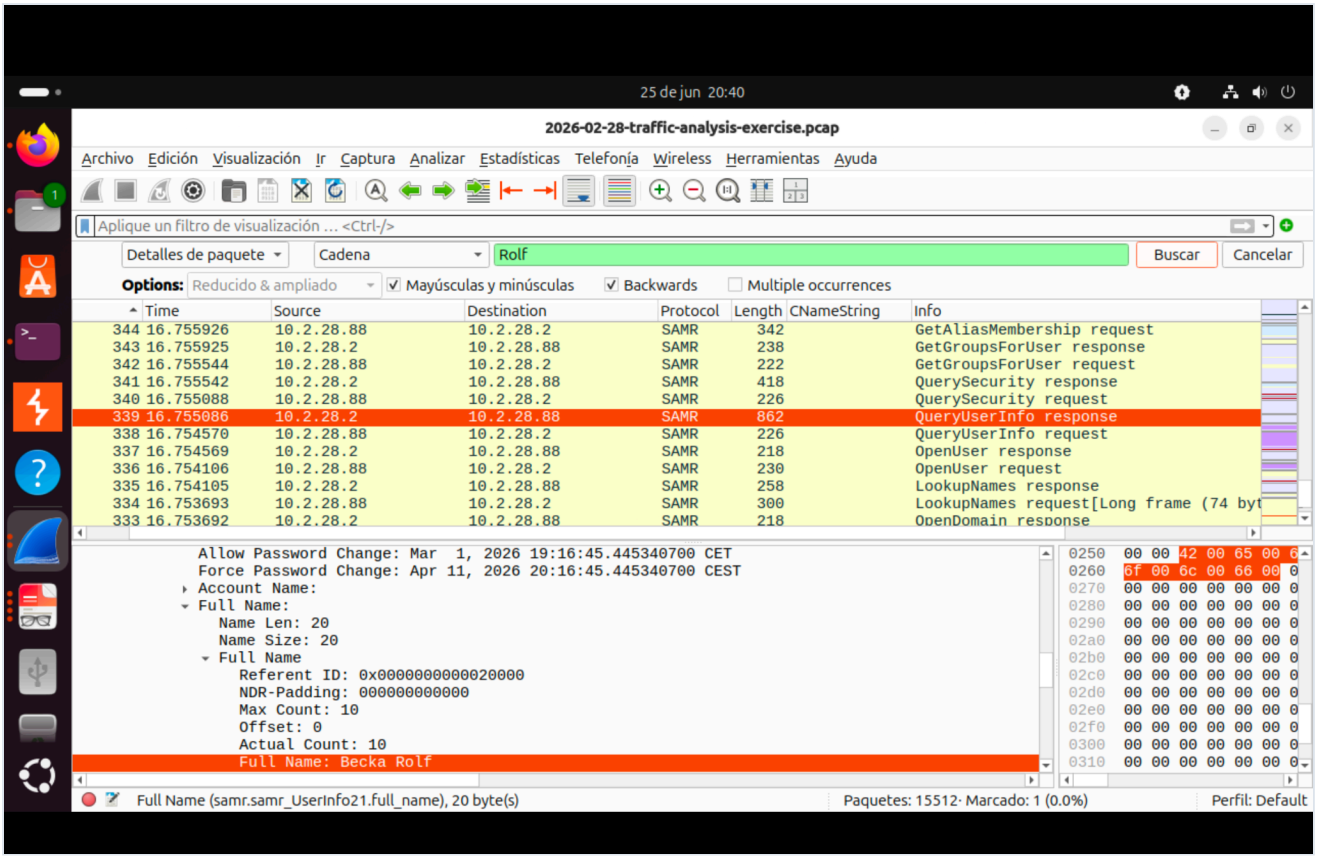
Packet List

That searches the packet list (or the selected column), rather than the packet's internal fields.

We made a mistake:

- First dropdown: Packet Details ← this is the one you still need to change.
- Second dropdown: String.
- Text: Rolf.
- Then click Find.

If “Packet Details” does not appear in that dropdown, tell me which version of Wireshark you are using (Help → About Wireshark), because some Ubuntu versions use a slightly different name for that option.



So now you have all five answers:

1. IP address: 10.2.28.88
2. MAC address: 00:19:d1:b2:4d:ad
3. Hostname: DESKTOP-TEYQ2NR
4. User account: brolf
5. Full name: Becka Rolf